GridIQ

Power-quality assessment

Demo Corporation · Sandton Mall

PROVISIONAL: the assessment is still running. Results cover the elapsed period only and are not final.

Assessment
GQ-A-0005 · 21-day assessment

Report
revision 0 · provisional preview

Observation period (SAST)
2026-11-01 21:00 -> 2026-11-12 07:26

Observed
10.43 days of 21

Site address
99 Retail St, Sandton

Measurement points
HVAC Feed (incomer)

Prepared by GridIQ from measured data. No engineer's sign-off is implied unless a named reviewer appears on this page.

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Executive assessment

Main measurement point: HVAC Feed. Period 2026-11-01 21:00 to 2026-11-12 07:26 SAST.

On weekdays the load is highest between 13:00 and 17:00 (median of 7 weekdays).

Outside operating hours the load falls to 42 % of its operating-hours level.

Voltage distortion is highest around 13:00, 14:00, 15:00 (phase A, B, C).

2 areas merit follow-up.

Key numbers

Energy imported
110 922 kWh

Maximum demand
968.1 kVA

Load factor
49 %

Areas to follow up
2 of 4

Energy imported: Register energy over the observed 250 h (mixed). Maximum demand: Highest 30-minute average (billing basis); contracted 1400 kVA. Load factor: Average kW ÷ maximum 30-minute kW. Typical operating load 660.5 kW. Areas to follow up: 2 prioritised observation(s); 0 area(s) with insufficient evidence.

Assessment by area

Area	Status	Basis
Measurement quality	Investigate	51 % of expected 10-minute slots measured (rule: investigate below 80 %).
Supply observations	Investigate	voltage THD P95 above 8 % on A, B, C (NRS 048-2 screening thresholds).
Installation and load behaviour	No material issue observed	Loaded PF 0.933, current unbalance P95 1.79 %, after-hours load 42 % of operating.
Demand exposure	No material issue observed	Maximum 30-minute demand 968.1 kVA, 69 % of the contracted 1400 kVA.

Immediate priorities

- 1. Voltage THD on phase A exceeded 8 % (95th percentile 9.48 %), worst around 13:00, 14:00, 15:00. Individual harmonic orders were not recorded, so the source cannot be identified from this data. Next: Log individual harmonic orders per phase, compare distortion with site operation, and request supply (PCC) information. (electrical engineer; medium confidence)
- 2. Only 51 % of the expected measurements were received; conclusions for this point are limited. Next: Restore the meter connection; GridIQ will check what data can be recovered. (GridIQ; high confidence)

Important measurement limits

- Observation period 10.4 days (2026-11-01 21:00 to 2026-11-12 07:26 SAST); results describe this period only. No annual figure is extrapolated.
- One-second RMS trends cannot prove that no sub-second disturbance occurred; events listed are inferred from sampled trends unless marked device-recorded.
- A single net grid meter cannot separate PV production, battery behaviour or individual equipment loads.

1. Scope and method

Ten-minute values of every measured phase across the observation period, screened against NRS 048-2 (screening thresholds; formal compliance needs a Class A instrument, one full week and >= 95 % valid 10-minute values). Observed values, derived indicators, rule findings, possible causes and recommended verification are kept separate below.

Point	Coverage	10-min slots measured	Samples	Reporting interval	Gaps >= 30 min
HVAC Feed	51.1 %	771 of 1 502	23 068	mixed: ~300 s, ~10 s	2

Coverage: unique 10-minute slots of the observation window; each slot counts min(1, samples × its own reporting interval ÷ 600 s); samples are unique per meter and timestamp (duplicates are rejected at ingest). Missing data is never filled or shown as zero.

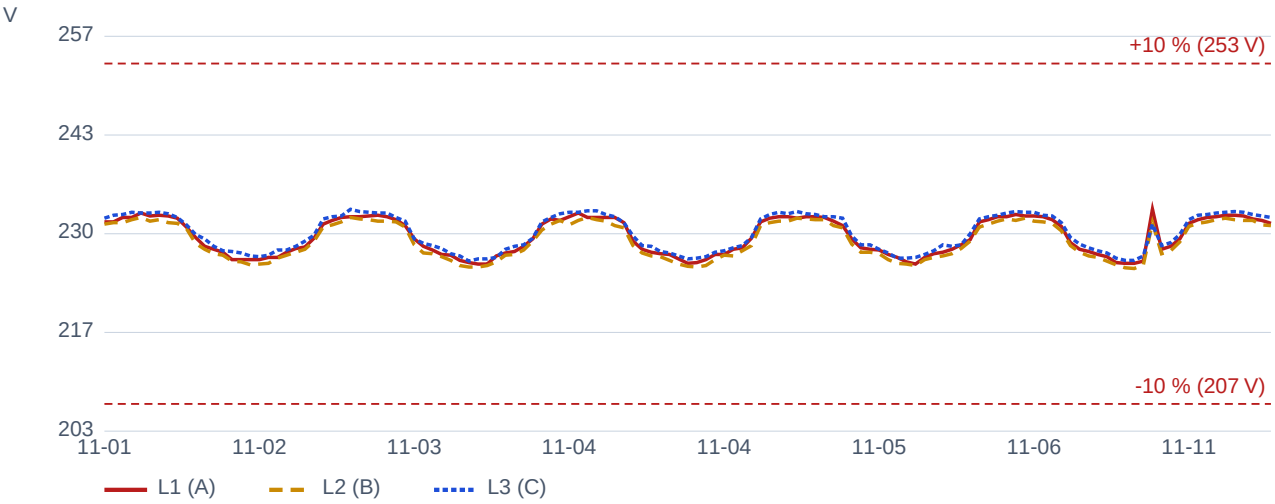
2. HVAC Feed

Verdict type: screening only. The meter is not confirmed as an IEC 61000-4-30 Class A instrument; values are screened against NRS 048-2 thresholds, not certified.

Voltage (observed)

Phase	Min (V)	Max (V)	P5 (V)	P95 (V)	Outside ±10 % of 230 V	10-min values
A	224,1	234,4	226	232,7	0.0 %	771
B	223,9	234,1	225,6	232,2	0.0 %	771
C	225,2	234,6	226,5	233,1	0.0 %	771

Figure 1. Phase voltage, hourly averages (SAST)

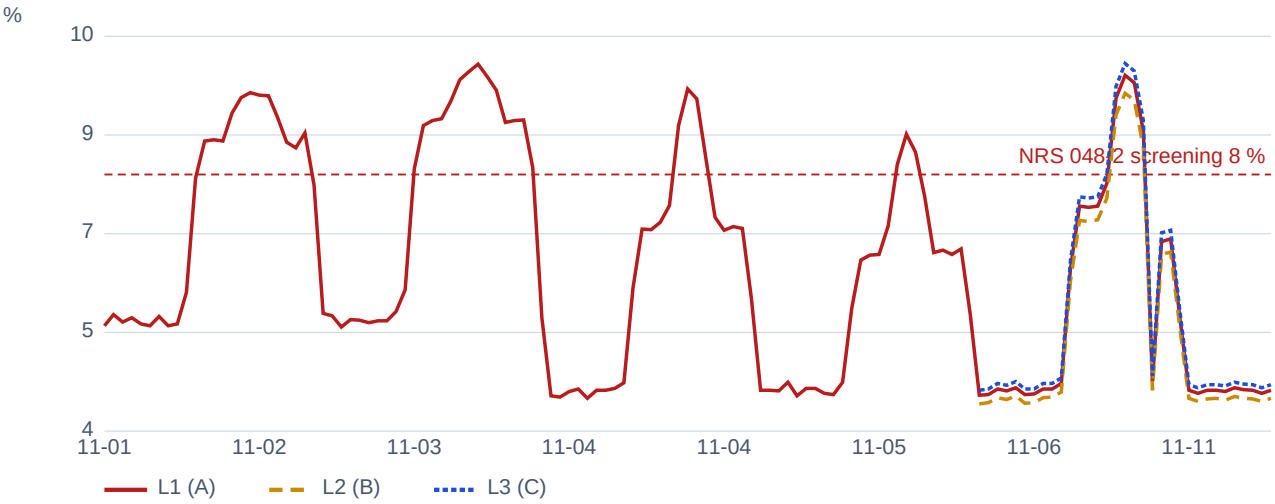


Hourly averages for readability; the table uses 10-minute values, so short excursions look smaller here.

Voltage distortion warrants investigation

Phase	THD-V P95 (%)	THD-V max (%)	10-min values > 8 %	Worst hours (SAST)
A	9.48	10.17	28.5 %	13:00, 14:00, 15:00
B	9.22	9.44	12.4 %	13:00, 14:00, 15:00
C	9.69	9.94	14.5 %	13:00, 14:00, 15:00

Figure 2. Voltage THD per phase, hourly averages (SAST)



Hourly averages; the exceedance shares in the table are counted on 10-minute values, not on these hourly points.

Current THD (P95): A 41.5 %, B 43.3 %, C 41.2 %. THD-I is relative to the present current, not TDD: no IEEE 519 conclusion is drawn without the demand current and fault level.

Individual orders: Individual harmonic orders were not recorded (THD only). No spectrum is shown or inferred. To identify harmonic sources, log individual orders (e.g. 3, 5, 7, 11, 13) per phase.

Possible causes (hypotheses, to verify)

- Non-linear loads on site: variable-speed drives, rectifiers, UPS, LED lighting. Distortion that follows the site's operating hours points this way.
- Background distortion from the supply network: distortion present outside operating hours, or at a similar level on all phases, points this way.
- Resonance with PF-correction capacitors (if installed and not detuned) can amplify a harmonic order.

Recommended follow-up measurements

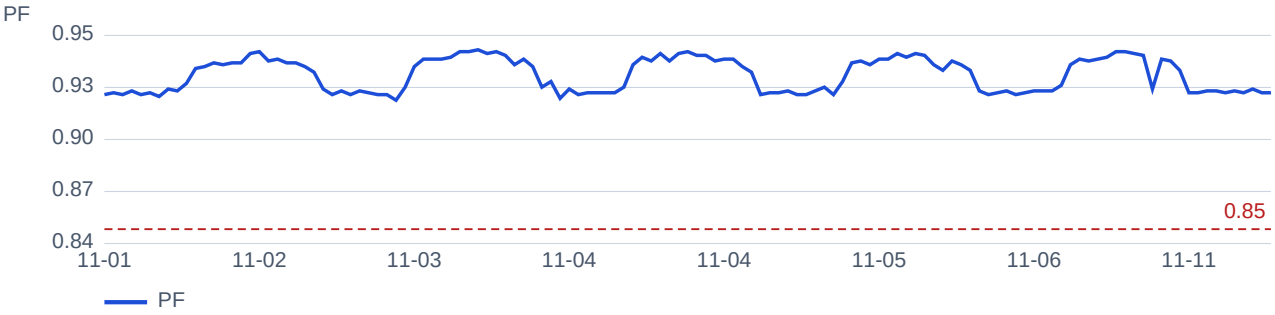
- Log individual voltage and current harmonic orders (at least 3, 5, 7, 11, 13) per phase for one week at the incomer and the main suspected feeders.
- Record the transformer rating and the fault level at the point of supply; list drives and capacitor banks with their ratings.

No filter or capacitor size is recommended from THD data alone; that needs an engineering study.

Power factor and reactive demand

Average power factor while loaded 0.933; 5th percentile 0.919; 0 % of loaded 10-minute values below 0.85. Low PF increases kVA and reactive charges; it is not a supply-quality breach.

Figure 3. Power factor, hourly averages (SAST)



Unbalance, frequency, interruptions and events

Voltage unbalance P95 0.44 %; current unbalance P95 1.8 % (unbalance as computed by the meter (per its configuration); not stated to be negative-sequence unbalance). Frequency 49.86 Hz–50.14 Hz; 0 10-minute value(s) outside 49–51 Hz. 0 minute(s) with voltage below 23 V.

0 excursion event(s). Source: inferred from the sampled RMS trend (pq_events), not device-recorded. device event/waveform capture not confirmed: transient and sub-second claims are not made.

Rule findings for this point

Finding	Observed	Recommended verification	Review
Voltage distortion above the NRS 048-2 limit (THD 9.7 %)	The 95th percentile of 10-minute voltage THD was 9.7 % on phase C, above the 8 % limit (last 7 days, 33.6 h metered). Sustained distortion can add heating in motors and transformers; whether it affects equipment here is not established by this data.	Log individual harmonic orders per phase, compare the distortion with site operation, and obtain supply (point of common coupling) information. Whether the source is on site or in the network is a hypothesis to verify; no filter is sized from THD alone.	not yet reviewed

Technical appendix

Dataset schema
gridiq.assessment-dataset/1

Dataset SHA-256
380d3b9aa96ec7cc128013b583130527…

Calculation version
gridiq-calc-2026.11.2

Rule version
gridiq-rules-2026.11.2

Template version
gridiq-templates-2026.11.1

Data cutoff (SAST)
2026-11-12 07:26

Generated (SAST)
2026-11-12 07:26

Time convention
[start, end): a value at bucket B covers B <= t < B + interval

Method

- Readings are attributed to a measurement point by deployment window at the measurement time; buckets that straddle an install or removal are recomputed from the samples inside the window.
- Energy from cumulative kWh register differences per deployment; a negative step is a register reset and never summed through; power integration only where registers are absent.
- Billed demand: 30-minute minute-weighted kVA averages; instantaneous peaks are reported separately.
- Power quality: 10-minute values screened against NRS 048-2 (screening thresholds; formal compliance needs a Class A instrument, one full week and >= 95 % valid 10-minute values).
- Sign convention: import positive; the deployment sign correction is applied. Times in the data are UTC; this report shows SAST (UTC+2).

Full SHA-256 of the dataset: 380d3b9aa96ec7cc128013b583130527022b7b826f7e58788453fb1bfe1ee655